

Machine Learning Based **Diabetes Prediction System**

1. Problem Statement

Diabetes is a chronic disease affecting millions of people worldwide. Early detection is critical to prevent complications like kidney failure, heart disease, and nerve damage. The goal of this project is to build a machine learning system that can predict whether a patient is diabetic or not, based on their clinical data.

2. Dataset

Name: Diabetes Prediction Dataset **Source:** Kaggle — [iammustafatz/diabetes-prediction-dataset](#) **Size:** 100,000 patient records **Features:** 8 input features + 1 target

Feature	Description
gender	Male / Female
age	Age in years
hypertension	Has hypertension or not (0/1)
heart_disease	Has heart disease or not (0/1)
smoking_history	never / former / current / unknown
bmi	Body Mass Index
HbA1c_level	Glycated haemoglobin — key diabetes marker
blood_glucose_level	Fasting blood glucose in mg/dL
diabetes	Target — 0 = Non-Diabetic, 1 = Diabetic

Known Issues with Dataset:

- Heavily imbalanced — only 8.5% diabetic, 91.5% non-diabetic
- Unverified source — no documented hospital or institution
- Smoking history had 6 ambiguous overlapping categories
- Only 18 rows had "Other" gender — too few to be useful

3. Why We Chose This Dataset

We initially used the PIMA Indians Diabetes Dataset but switched because:

- PIMA has only 768 rows — too small
- PIMA covers only females of one ethnic group — not generalisable
- PIMA has biologically impossible zero values in glucose, insulin, BMI
- Our mentor asked us to switch to a better dataset

We then evaluated 4 datasets and chose iammustafatz because:

- 100,000 rows — large enough for robust training
 - Contains HbA1c and blood glucose — real clinical diagnostic features
 - Actively used in published research — citable
 - Suitable for demonstrating full ML pipeline
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4. ML Pipeline — Step by Step

Phase 1 — Exploratory Data Analysis

- Plotted class distribution — confirmed 8.5% vs 91.5% imbalance
- Plotted histograms for all numeric features
- Drew correlation heatmap — HbA1c and blood glucose most correlated with target
- Drew box plots — clear separation between diabetic and non-diabetic patients for HbA1c and glucose

Phase 2 — Data Preprocessing

- **Smoking history** — merged 6 messy categories into 4 clean ones (never, former, current, unknown). "ever" and "not current" were merged into "former" as they were ambiguous
- **Gender** — removed 18 rows with "Other" gender — too few to affect model
- **Duplicates** — dropped all duplicate rows
- **Label Encoding** — converted gender and smoking_history to numbers
- **Standard Scaling** — scaled all features to mean=0, std=1 using StandardScaler

Phase 3 — Train/Test Split + SMOTE

- Split dataset 80% training, 20% test using stratified split — preserves class ratio in both sets
- Applied **SMOTE on training data only** — this is critical
- SMOTE balanced the training set from 8.5%/91.5% → 50%/50%
- Test set was never touched — kept at real-world 8.5% diabetic distribution
- Training set grew from 79,985 rows → 146,374 rows after SMOTE

Why SMOTE on training only? Applying SMOTE on the full dataset before splitting causes **data leakage** — synthetic samples derived from training points leak into the test set, making evaluation metrics artificially inflated. This is a documented mistake. The correct approach is always split first, then SMOTE on training only.

Phase 4 — Feature Selection

- Ran a preliminary Random Forest to get feature importance scores
- Checked correlation of each feature with the target
- Decision: **retained all 8 features** — small feature set, all clinically justified, no dimensionality issue

Phase 5 — Model Training

Trained 5 models and evaluated all on the same held-out test set:

Model	Type
Logistic Regression	Linear baseline
Random Forest	Bagging ensemble
XGBoost	Boosting ensemble
Soft Voting	RF + XGBoost vote on probabilities
Stacking	RF + XGBoost → Logistic Regression meta-model

Why these 5?

- Logistic Regression as a simple interpretable baseline
- Random Forest and XGBoost are themselves ensemble methods
- Soft Voting and Stacking to check if combining models improves performance

Evaluation metrics used:

- ROC-AUC — primary metric
- F1 Score — balances precision and recall
- Precision and Recall — per class
- Confusion Matrix — shows false negatives (missed diabetics)
- We deliberately avoided using accuracy alone — misleading with imbalanced data

Phase 5 — Results

Model	ROC-AUC	F1	Precision	Recall
XGBoost ★	0.9757	0.8003	0.8908	0.7264
Soft Voting	0.9749	0.7903	0.8450	0.7423
Stacking	0.9741	0.7775	0.8071	0.7500
Random Forest	0.9692	0.7587	0.7545	0.7630
Logistic Regression	0.9616	0.5759	0.4265	0.8862

XGBoost won — highest ROC-AUC on test set.

Why did ensemble not beat XGBoost? Ensembling works when models make different mistakes and cover each other's blind spots. In this dataset, HbA1c and blood glucose make the answer so obvious that all models learn the same pattern. XGBoost, being a boosting ensemble of 100 trees, already captures the maximum learnable signal. Adding Random Forest through voting introduced slight noise rather than complementary signal — marginally reducing AUC.

Phase 6 — SHAP Explainability

SHAP (SHapley Additive exPlanations) was applied to explain why the model makes each prediction.

SHAP Feature Importance Ranking:

1. HbA1c_level — dominant predictor
2. blood_glucose_level — dominant predictor
3. age — moderate importance
4. bmi — moderate importance

5. hypertension — low importance
6. heart_disease — low importance
7. smoking_history — very low importance
8. gender — near zero importance

What this proves:

- Model learned exactly what doctors learn — HbA1c and glucose are the clinical definition of diabetes
- SHAP dependence plots show the model's threshold matches clinical thresholds — $\text{HbA1c} \geq 6.5\%$ and $\text{blood glucose} \geq 140 \text{ mg/dL}$
- Gender near zero importance hints the dataset may not be fully real clinical data — acknowledged as a limitation
- This is not overfitting — the model learned medically valid logic, not noise

Cross Validation:

- 5-fold stratified cross validation on XGBoost
- All 5 fold AUC scores consistent with low standard deviation (< 0.02)
- Proves model generalises well — not memorising training data

5. Overfitting Analysis

Check	Result
Train AUC	0.9982
Test AUC	0.9757
Gap	0.0225 — acceptable
CV Std Dev	< 0.02 — consistent

SHAP top features HbA1c + glucose — medically valid

Verdict: Not overfitting. The high AUC is because HbA1c and blood glucose are literally the medical definition of diabetes — the model learned a clinically valid threshold rule, not random noise.

Saved files:

- best_diabetes_model.pkl — XGBoost model
 - scaler.pkl — StandardScaler
 - le_gender.pkl — Gender LabelEncoder
 - le_smoking.pkl — Smoking LabelEncoder
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6. Key Jury Defence Points

Question	Answer
Why SMOTE on training only?	Applying on full data causes data leakage — synthetic samples leak into test set, inflating metrics
Why is AUC so high?	HbA1c and glucose are clinical definition of diabetes — strong signal, not overfitting
Is it overfitting?	No — cross-val std dev < 0.02, train-test gap 0.0225, SHAP shows medically valid features
Why did ensemble not win?	All models learn the same pattern — XGBoost already captures maximum signal as a 100-tree boosting ensemble
Why not use accuracy?	Dataset is 91.5%/8.5% imbalanced — a model predicting "no diabetes" always gets 91.5% accuracy, which is clinically useless
Dataset limitation?	Unverified source, possible synthetic data, gender near zero SHAP — all acknowledged honestly

8. Tools and Technologies

Tool	Purpose
Python	Core language
Google Colab	Training environment
Pandas / NumPy	Data handling
Scikit-learn	Preprocessing, models, metrics

Tool	Purpose
XGBoost	Final prediction model
imbalanced-learn	SMOTE
SHAP	Explainability
Matplotlib / Seaborn	Visualisations
Streamlit	Web application
Joblib	Model saving